

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1240

Hamilton Peaking 100 kV
November 20, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1240. The Interconnection Customer proposes a 106.2 MW solar photovoltaic facility interconnecting at the Hamilton Peaking 100 kV substation in the Ohio area. The request was received on July 09, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 408 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,232,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Four Corners to Lantzville 100kV is overloaded to 110.1% of its 140 MVA emergency rating for the loss of the Hamilton Peaking to J B Sims 100kV. The project contributes 11.4% DFAX (12.1 MW) to this overload.

For the loss of Four Corners to Lantzville 100kV, the J B Sims to Hamilton Peaking 100kV line loads to 102.4% of its 169 MVA rating. Project DFAX: 19.2%.

For the loss of Hamilton Peaking to J B Sims 100kV, the Memorial to Lantzville 100kV loads to 101.9% of its 133 MVA rating. Project DFAX: 11.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Four Corners to Lantzville 100kV	Rebuild line with higher-capacity conductor	\$2,439,000	33% (\$799,000)
J B Sims to Hamilton Peaking 100kV	Reconductor line	\$161,000	54% (\$86,000)
Lantzville to Memorial 100kV line	Reconductor line	\$1,060,000	33% (\$347,000)
	Total allocated to project		\$1,232,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.